

Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить перенаправление ввода-вывода
- 2 Изучить работу фильтров
- 3 Изучить команду поиска
- 4 Ознакомиться с управлением процессами
- 5 Ознакомиться с командами `df` `du`

2. Процесс выполнения лабораторной работы

```
vdmorozov@vdmorozov:~$ ls /etc/ > file.txt
vdmorozov@vdmorozov:~$ ls >> file.txt
vdmorozov@vdmorozov:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
```

Рисунок 1: Запись в файл

```
vdmorozov@vdmorozov:~$ grep .conf file.txt > conf.txt
vdmorozov@vdmorozov:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprind.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
libaudit.conf
```

Рисунок 2: Поиск расширения .conf


```
/home/vdmorozov/.npm/_cacache/index-v5/c0  
/home/vdmorozov/.npm/_cacache/index-v5/09/c0  
/home/vdmorozov/.npm/_cacache/index-v5/09/c0/cce94e2c664a8f390a8f1ade3346d5c95f3c  
ed5cf6fc29def30220242c53  
/home/vdmorozov/.npm/_cacache/index-v5/59/e7/cb225f160e685933e8439d3cf624ef721a58  
996cfad55fb2000f59e2edcc  
/home/vdmorozov/.npm/_cacache/index-v5/cd  
/home/vdmorozov/.npm/_cacache/index-v5/44/44/c9a2c5dc2999cbe868c370ef479e17dfd1e8  
6117d1350482a1d990504757  
/home/vdmorozov/.npm/_cacache/index-v5/c0  
/home/vdmorozov/.npm/_cacache/index-v5/c9  
/home/vdmorozov/.npm/_cacache/index-v5/9f/ae/c077854801f7b4b5cf7eea2bb68ca2bc53f0  
b53cea310cda0b203217f693  
/home/vdmorozov/.npm/_cacache/index-v5/af/cf  
/home/vdmorozov/.npm/_cacache/index-v5/c3  
/home/vdmorozov/.npm/_cacache/index-v5/a1/5e/c9aa1c46bf0c67511379e97bd07fca875324  
22f2506214f94db8861160d3  
/home/vdmorozov/.npm/_cacache/index-v5/78/34/c1e131014ad3a4ee13f096e70f74c03e6883  
7bc2c8188a44d554f3f3ba2e  
/home/vdmorozov/.npm/_cacache/index-v5/ca  
/home/vdmorozov/snap/hugo/common  
/home/vdmorozov/snap/hugo/current  
/home/vdmorozov/conf.txt  
vdmorozov@vdmorozov:~$
```

Рисунок 3: Поиск файлов

```
find: '/etc/lvm/archive': Permission denied
find: '/etc/lvm/backup': Permission denied
find: '/etc/lvm/cache': Permission denied
find: '/etc/lvm/devices': Permission denied
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
find: '/etc/openvpn/server': Permission denied
/etc/foomatic/hashes.d
/etc/foomatic/hashes.d/hashes.new
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
/etc/logrotate.d/httpd
/etc/nvme/hostnqn
:find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
find: '/etc/ssh/sshd_config.d': Permission denied
find: '/etc/sss.d': Permission denied
find: '/etc/sudoers.d': Permission denied
find: '/etc/userdb': Permission denied
```

Рисунок 4: Поиск файлов

```
vdmorozov@vdmorozov:~$  
vdmorozov@vdmorozov:~$ find /etc -name "h*" -print | less  
find: '/etc/audit': Permission denied  
vdmorozov@vdmorozov:~$  
vdmorozov@vdmorozov:~$  
vdmorozov@vdmorozov:~$ find ~ -name "log*" > logfile &  
[1] 6956  
vdmorozov@vdmorozov:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
vdmorozov@vdmorozov:~$ rm logfile  
vdmorozov@vdmorozov:~$
```

Рисунок 5: Фоновый запуск процесса

```
vdmorozov@vdmorozov:~$  
vdmorozov@vdmorozov:~$ gedit &  
[1] 6995  
vdmorozov@vdmorozov:~$ ps | grep gedit  
    6995 pts/1    00:00:00 gedit  
vdmorozov@vdmorozov:~$ kill 6995  
[1]+  Terminated                  gedit  
vdmorozov@vdmorozov:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

```
DF(1)                                User Commands                                DF(1)

NAME
    df - report file system space usage

SYNOPSIS
    df [OPTION]... [FILE]...

DESCRIPTION
    This manual page documents the GNU version of df.  df displays the amount
    of space available on the file system containing each file name argument.
    If no file name is given, the space available on all currently mounted
    file systems is shown.  Space is shown in 1K blocks by default, unless
    the environment variable POSIXLY_CORRECT is set, in which case 512-byte
    blocks are used.

    If an argument is the absolute file name of a device node containing a
    mounted file system, df shows the space available on that file system
    rather than on the file system containing the device node.  This version
    of df cannot show the space available on unmounted file systems, because
    on most kinds of systems doing so requires non-portable intimate knowl-
    edge of file system structures.

OPTIONS
    Show information about the file system on which each FILE resides, or all

Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

```
DU(1)                                User Commands                                DU(1)

NAME
    du - estimate file space usage

SYNOPSIS
    du [OPTION]... [FILE]...
    du [OPTION]... --files0-from=F

DESCRIPTION
    Summarize device usage of the set of FILES, recursively for directories.

    Mandatory arguments to long options are mandatory for short options too.

    -0, --null
        end each output line with NUL, not newline

    -a, --all
        write counts for all files, not just directories

    --apparent-size
        print apparent sizes rather than device usage; although the apparent size is usually smaller, it may be larger due to holes in ('sparse') files, internal fragmentation, indirect blocks, and the like

Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

```
vdmorozov@vdmorozov:~$  
vdmorozov@vdmorozov:~$ df  
Filesystem      1K-blocks    Used Available Use% Mounted on  
/dev/nvme0n1p3 155186176 91903640 62709592 60% /  
devtmpfs        4011280      0    4011280  0% /dev  
tmpfs           4035652     100    4035552  1% /dev/shm  
tmpfs           1614264     1960    1612304  1% /run  
tmpfs           1024         0      1024    0% /run/credentials/systemd-journald.service  
tmpfs           4035656     4096    4031560  1% /tmp  
/dev/nvme0n1p3 155186176 91903640 62709592 60% /home  
/dev/nvme0n1p2 1992552     518800    1352512 28% /boot  
/dev/loop1      115712     115712         0 100% /var/lib/snapd/snap/hugo/25676  
/dev/loop2      49280      49280         0 100% /var/lib/snapd/snap/snapd/25935  
/dev/loop0      68480      68480         0 100% /var/lib/snapd/snap/core24/1349  
tmpfs           1024         0      1024    0% /run/credentials/systemd-resolved.service  
tmpfs           807128      176     806952  1% /run/user/1077  
/dev/sr0        2677920    2677920         0 100% /run/media/vdmorozov/Fedora-WS-Live-43  
vdmorozov@vdmorozov:~$
```

Рисунок 9: Справка по команде du

```
4      ./npm/_cacache/index-v5/2b
4      ./npm/_cacache/index-v5/30/1a
4      ./npm/_cacache/index-v5/30
424    ./npm/_cacache/index-v5
115940 ./npm/_cacache
116004 ./npm
12     ./snap/hugo/25676/.config/go/telemetry/local
0      ./snap/hugo/25676/.config/go/telemetry/upload
12     ./snap/hugo/25676/.config/go/telemetry
12     ./snap/hugo/25676/.config/go
12     ./snap/hugo/25676/.config
12     ./snap/hugo/25676
0      ./snap/hugo/common
16     ./snap/hugo
16     ./snap
624    ./texlive2025/texmf-var/luatex-cache/generic/names
90580  ./texlive2025/texmf-var/luatex-cache/generic/fonts/otl
90580  ./texlive2025/texmf-var/luatex-cache/generic/fonts
91204  ./texlive2025/texmf-var/luatex-cache/generic
91204  ./texlive2025/texmf-var/luatex-cache
91204  ./texlive2025/texmf-var
91204  ./texlive2025
1064684 .
vdmorozov@vdmorozov:~$
```

Рисунок 10: Запуск команды du


```
/home/vdmorozov/.npm/_cacache/index-v5/9c
/home/vdmorozov/.npm/_cacache/index-v5/9c/e7
/home/vdmorozov/.npm/_cacache/index-v5/2b
/home/vdmorozov/.npm/_cacache/index-v5/2b/8e
/home/vdmorozov/.npm/_cacache/index-v5/30
/home/vdmorozov/.npm/_cacache/index-v5/30/1a
/home/vdmorozov/snap
/home/vdmorozov/snap/hugo
/home/vdmorozov/snap/hugo/25676
/home/vdmorozov/snap/hugo/25676/.config
/home/vdmorozov/snap/hugo/25676/.config/go
/home/vdmorozov/snap/hugo/25676/.config/go/telemetry
/home/vdmorozov/snap/hugo/25676/.config/go/telemetry/local
/home/vdmorozov/snap/hugo/25676/.config/go/telemetry/upload
/home/vdmorozov/snap/hugo/common
/home/vdmorozov/.texlive2025
/home/vdmorozov/.texlive2025/texmf-var
/home/vdmorozov/.texlive2025/texmf-var/luatex-cache
/home/vdmorozov/.texlive2025/texmf-var/luatex-cache/generic
/home/vdmorozov/.texlive2025/texmf-var/luatex-cache/generic/names
/home/vdmorozov/.texlive2025/texmf-var/luatex-cache/generic/fonts
/home/vdmorozov/.texlive2025/texmf-var/luatex-cache/generic/fonts/otl
vdmorozov@vdmorozov:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.